



SRIKANT DATAR
SALONI CHATURVEDI
CAITLIN N. BOWLER

BMVSS: Changing Lives through Innovation One Jaipur Limb at a Time (Abridged)

People who live inside the world's many war zones from Afghanistan to Rwanda may never have heard of New York or Paris, but they are likely to know of a town in northern India called Jaipur. Jaipur is famous in strife-torn areas as the birthplace of an extraordinary prosthesis or artificial limb known as the Jaipur Foot that has revolutionized life for millions of landmine amputees. The beauty of the Jaipur Foot is its lightness and mobility – those who wear it can run, climb trees and pedal bicycles – and its low price.

– Tim McGirk, correspondent, *TIME*¹

In 1969, Devendra Raj Mehta (D.R. Mehta), who would later become chairman of the Securities and Exchange Board of India (SEBI), the Indian securities regulator, suffered serious injuries and almost lost his leg because of a motorcycle accident. Deeply moved by this experience, he established BMVSS in 1975 at Jaipur,^a as a non-governmental organization that would rehabilitate persons with physical disabilities by providing them with prosthetic limbs and other mobility-assisting devices, free of cost. Founded on the ethos that by helping people regain their mobility, it was enabling them to regain their livelihood and dignity, BMVSS functioned via a combination of grants and donations, and had assisted over a million people since its inception.

BMVSS was widely known throughout India and beyond for its contribution in improving and popularizing the legendary low-cost prosthesis, the Jaipur Foot, and in developing the Stanford-Jaipur Knee – an artificial knee joint that *TIME* had named one of the 50 best inventions of 2009, while also recognizing D.R. Mehta's efforts over the past 37 years. The organization proudly treated rich and poor patients alike, growing and sustaining operations through a unique combination of

^a Jaipur, the largest city in the western state of Rajasthan, is known for its colorful handicrafts and historical architecture.

Professor Srikant Datar and Research Associate Saloni Chaturvedi prepared the original version of this case "BMVSS: Changing Lives, One Jaipur Limb at a Time," HBS No. 114-007. This abridged version was prepared by Professor Srikant Datar and Research Associates Saloni Chaturvedi and Caitlin N. Bowler. Funding for the development of this case was provided by Harvard Business School and not by the company. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

government support, a range of donations from patients, corporate partners, and philanthropists, and several self-sustaining smaller enterprises.

How it all began

BMVSS' inception in 1975 was closely linked to the invention of the Jaipur Foot—a prosthesis for both below and above-the-knee amputees—in 1968.

Genesis of the Jaipur Foot

The Jaipur Foot's story began with a chance meeting between Ram Chandra Sharma, a sculptor adept at recreating human likeness, and Dr. Pramod Karan Sethi, an orthopedic surgeon. During his occupational therapy classes for polio-afflicted children at the Swai Man Singh (SMS) hospital in Jaipur, Sharma saw Sethi working with accident victims who had lost their limb(s). He observed that few patients were fitted with artificial limbs—the SMS Hospital produced only five or six such limbs per year based on American and German designs, as it was a time and skill-intensive process, and imported limbs were expensive. Sharma also noticed that there was limited acceptance for such limbs as they did not permit much mobility and could only be used with an attached shoe.

Dr. Pooja Mukul, technical consultant at BMVSS, described the situation,

Although these artificial limbs were working well in the West, they were found inappropriate for Indian amputees. This was largely on account of the vocational and socio-cultural needs for which the amputees had to squat, sit cross-legged, negotiate rugged terrain, walk barefoot, etc. None of these postures and activities were a need of the western amputee. In addition, everybody in the West wears shoes, so the cosmetic appearance of the foot was not a consideration as it would be concealed within the shoe. In India a lot of people do not wear shoes during activities such as visiting places of worship and the foot is often exposed. Therefore it is imperative that it be durable, waterproof and look like a normal human foot, so that it can be used with or without a shoe depending on the amputee's need.

Sharma, vexed by the pain and discomfort that the artificial limbs caused Sethi's patients, was convinced that he could use his sculpting skills to model an artificial foot that looked and functioned more like a human foot. He approached Sethi with his idea and over the next two years, the two along with Dr. S.C. Kasliwal and Dr. Mahesh Udawat from the SMS Hospital, worked on developing several alternatives. However, their material choices made the resulting limbs too flimsy or bulky. Apart from trying to design a flexible foot, the three doctors and Sharma focused on socio-cultural needs. Most amputees at the SMS Hospital were poor and engaged in physical labor. Hence, the loss of a limb affected their livelihood. They required a low-cost prosthesis that could be manufactured and fitted quickly—using a simple process and locally available materials.

One day, while getting his bicycle's flat tire repaired, Sharma noticed a mechanic retreading a truck tire with vulcanized rubber.² This sparked an idea and Sharma requested the mechanic to cast a foot using vulcanized rubber. Although the resulting foot was more flexible than earlier models, it shredded a few days later.

The doctors and Sharma had also been working on refining the Solid Ankle Cushion Heel (SACH) foot, a Western design that consisted of a rigid wooden block covered in rubber (see **Exhibit 1**). As

the wooden block ran from ankle to instep, it permitted almost no flexibility. While tinkering with its design, the duo took wedges off, converting the single wooden block into two to increase flexibility.

Merging the idea of the vulcanized rubber foot and the re-worked SACH foot, the three doctors and Sharma wrapped the separated wooden block in light rubber and vulcanized the resulting structure to form the first successful design of the Jaipur Foot—the only non-articulated (without moving joints) artificial foot that allowed several types of movements. For instance, the Jaipur Foot could flex during squatting or climbing slopes and rotate inward while sitting cross-legged. To make the foot wearable for both above and below-the-knee amputees, either, a shank^b and brace;^c or a shank, brace, and knee joint, were used to connect the foot to the patient's limb (see **Exhibit 2**). At this time, production cost of the Jaipur Foot was less than \$5.

Further improvements led to a design with three pieces—wooden ankle piece, sponge rubber forefoot, and heel—that were wrapped in rubber and vulcanized in an aluminum casting die (see **Exhibit 3**). Mukul described the process, “It wasn’t research in a very formal sense, but it was all based on user feedback. So the patient-centric approach started from the beginning, it was all about what the patients want, how they want it, what they approve of, what is appropriate for them.” The Jaipur Foot won a lot of international recognition as it found use among landmine and war victims.

Vinod Rawat, a patient who had lost his foot as a child, described the experience of being fitted with the prosthesis, “One hour later, I was walking without crutches. Now I run the Mumbai Marathon, I climb mountains, I ride motorbikes.”³

The Organization: More than just Brick and Mortar

Next to God, I respect the people [at BMVSS] who have given me my life back. Everyone in my village said I would never walk again. I could have ended up begging at the traffic lights.

— Danny Vithal , a former patient who went on to work at BMVSS⁴

Despite being a low-cost breakthrough in prosthetics, only 50 patients were fitted with the Jaipur Foot during 1968 to 1975. Once BMVSS actively took up fitment of limbs, this number grew to 109 in the following year, and gradually increased to 10,000 by 1982. Under the auspices of an executive committee and with the assistance of its 200-odd staff members, by 2013, BMVSS through its Jaipur center, other branches and outreach camps was fitting, on an average, close to 22,000 limbs annually. It was also providing assistance in the form of calipers,^d crutches, tricycles, wheelchairs and hearing aids, to over 40,000 others (see **Exhibit 4**).

D.R. Mehta took pride in the fact that BMVSS had different operating practices compared to other healthcare facilities, welcoming and treating both poor and rich patients, alike. All patients could walk into the organization's premises at any time, without prior appointment or registration. Even the guard could admit patients and a doctor would then examine them during the next set of working hours. D.R. Mehta commented, “Poor people... do not know how to read and write. How do you expect them to write to a hospital for an appointment or register themselves?” He lamented the

^b Shank was the part of the prosthetic limb that connected the knee and ankle. One end of the shank was connected to the Jaipur Foot and the other to a total contact socket which seamlessly fitted the patient's limb.

^c Brace was a device used to fasten the shank to the patient's limb.

^d Calipers were mobility assisting prosthesis used by patients affected by polio.

fact that while providing quality care, private healthcare systems often excluded a section of society. Expressing his despair, D.R. Mehta said, “Today, the healthcare models are hospital or doctor-centric, and based on the concept of user charges with the result that the neediest get excluded, even in a country like ours. There is a perception that money is the measure of everything in life and that if patients do not pay, then they do not appreciate the importance of the care given to them.”

While the Government of India (GoI) had, just like in many other countries, created a state-supported national healthcare system to address this problem of access, quality of care was not adequate. In the face of this, BMVSS presented a new patient-centric, corpus and donor fuelled model of quality care. This raised an interesting issue of whether such a model was scalable and replicable across the entire spectrum of healthcare services.

Although BMVSS did not advertise its work, except during camps, people from across India flocked to its Jaipur center based on word-of-mouth publicity. On a given day, one could find people who had travelled from far-flung states like Kashmir in the north and Kerala in the south. D.R. Mehta explained, “Every auto-driver^e at the station knows that if a person without a limb alights from the train, he is headed to BMVSS. The drivers don’t need an address; they know where to bring them.”

BMVSS provided meals and accommodation at no charge to patients and accompanying family members. The lean organization structure and assembly line-approach to limb fitment helped BMVSS fit an average of 40 to 50 patients daily. A patient could usually be fitted with a limb in as little as three hours. Barring few complex and time-consuming cases, most patients returned home in one to three days. If a patient was dissatisfied, the technician and attending doctor would work on the fitment until the patient was happy with the prosthesis. In fact, patients could be quite critical, sometimes returning after weeks, demanding that the fitting be adjusted to redress their discomfort. Staff had been trained to seek patient feedback to ensure continuous improvement.

In keeping with its focus on social rehabilitation of patients, BMVSS had partnered with a government-run vocational training center, to which it referred patients for training in skills such as carpentry and stitching. BMVSS also provided aids like sewing machines and tea-stall kits that could help patients become self-employed. D.R. Mehta even insisted that poor patients leaving the center were provided a set of new clothes. Often, during his evening rounds, he would give patients money for return train fare or other financial needs.

Another special characteristic of the organization was that a number of technicians (15% of total staff) were former patients, who had been trained in the Jaipur limb technology. Technicians and artisans who could empathize with patients, having gone through a fitment themselves, helped strengthen BMVSS’s patient-centric ethos. Dr. A.B. Sharma, workshop in-charge, said, “There are times when several patients want to be fitted the very day they arrive. It would be very easy for our staff to say, ‘No sir, I’ll not have that much time today.’ However, they are committed to serving the patients and try to accommodate as many fitments as possible.”

Outreach

In addition to its main center at Jaipur, BMVSS had established 22 branches across India. Although each branch operated independently, it had to mandatorily use the Jaipur Foot technology and comply with the guidelines set by BMVSS’s executive committee. Branches were usually set up with local support and some had even been set up in local hospitals.

^eAutos were three-wheeled vehicles used for public transport in many cities in India.

BMVSS also held regular camps, both within India and internationally. On an average, it conducted 50 camps each year. Since its inception, BMVSS had organized over 50 camps across 26 countries such as Afghanistan, Pakistan, Iraq, the Philippines, and Rwanda. Doctors and technicians usually travelled with the equipment to campsites, sometimes setting up a mobile unit to reach remote areas. BMVSS trained people at such sites on how to fabricate the foot and the fitment technique, leaving behind the equipment, so that fitments could continue even after the camp ended.

Financing

I am the biggest beggar in India.

— D.R. Mehta, on gathering donations for BMVSS

A significant portion of BMVSS's funding came from GoI and a mix of large donors such as Dow Chemical International Pvt. Ltd. (Dow), Paul Hamlyn Foundation and Glenmark Pharmaceuticals Ltd. (see **Exhibit 5**). Smaller one-time or recurring donations and grants, as well as interest from the corpus, made up the remainder.

Although both Dow and Glenmark were initially associated with BMVSS in different capacities, they eventually turned donors. Dow had started by experimenting with polyurethane as an alternative material for the Jaipur Foot and in 2006, began donating \$200,000^f to BMVSS annually. Glenmark became aware of BMVSS's work through the presence of D.R. Mehta on its Board of Directors. Soon after Glenmark's chairman visited BMVSS, the company offered BMVSS an annual donation of \$120,000.

BMVSS had several other sources of income. It had established a physiotherapy clinic on its premises, which charged one-fifth of the market rate and generated approximately \$40,000 as annual revenue—of this, around \$10,000 accrued as profit to BMVSS. It had also constructed shops on land it owned, which yielded an income of \$40,000. Over the years BMVSS had developed a reliable income stream.

Given the challenges it faced, some outside the organization questioned whether the model of relying solely on the corpus, grants and donations while not charging its patients for the limb or the fitment, could ensure that BMVSS would continue to serve the thousands it did annually.

D.R. Mehta understood the concerns, but felt very strongly on the issue, arguing, "The minute you start instituting a charge for your service, the most vulnerable populations are the first sector of society to be marginalized, and it's precisely this sector of the society that you wish to serve the most."

Bhupendra Raj Mehta, Joint Secretary, BMVSS, added, "We can have fundraisers or get a professional institution to help us collect money. However, we welcome everyone and are not concerned with whether the patient is a rich man or a poor man. The moment we start becoming concerned, we will be wasting our time and energy in separating them."

^f An exchange rate of \$1=₹50 has been used throughout the case.

Further Developments in Jaipur Limb Technology

Development of the Shank

Initially, the Jaipur Foot was fitted onto a below-the-knee amputee using an aluminum shank to connect the patient's limb and foot. To make the shank, technicians would heat an aluminum sheet to very high temperatures to make it malleable, roll it into a cone, weld the cone, insulate it and shape it according to the patient's leg. The process was labor-intensive and required technicians to be highly skilled. Also, the cone needed to be covered with a skin-colored stocking to resemble a human limb.

In 1985, during a visit to a businessperson's residence, Sharma saw a hollow black pipe hung on a wall. On being questioned, the businessperson explained that this was an irrigation pipe made of high-density polyethylene (HDPE). Since HDPE was sturdy, durable and could be molded when hot, Sharma thought there was a possibility of using the pipe instead of the aluminum cone, to make the shank. He then worked with BMVSS's research team to develop the technology needed to craft the shank out of the HDPE pipe. The use of the pipe was successful as it got rid of a potential stress point, the seam—that is, the part where the aluminum cone was welded together. Technicians oven-heated the pipes and molded them according to each patient's limb measurement. Since shaping the heated HDPE pipe was easier than shaping and welding the aluminum cone, even technicians with basic training could undertake the fitment.

The pipes offered another advantage—Mukul explained, “Because we do a lot of technology transfer, the simpler the process, the easier it is to transfer.” The pipes also catered to children's needs, as they were available in smaller diameters. Once BMVSS began putting the pipes to large-scale use, the manufacturer added a skin-colored pigment to the pipe.

The Stanford-Jaipur Knee

The knee joint BMVSS used for above-the-knee amputees was a single axis joint that operated like a door hinge. Despite allowing bending and straightening of the leg, this knee required extra effort in moving from the neutral or straight position to the flexed or bent position. According to Mukul, “The knee has a paradoxical function; when you are standing, you want the knee to be stable and not buckle, but when you want to move forward, you want it to bend. When you have a single axis joint, this axis is fixed behind the natural axis for stability while standing. However, since the axis is too far behind, when you bend the knee to take it forward, it requires extra energy.”

To overcome this issue, Dr. Armand Neukermans a well-known scientist and entrepreneur, who was an alumnus of Stanford University arranged for discussions between D.R. Mehta and Professors Thomas Andriacchi, Paul Yock and others at Stanford University. Stanford University and BMVSS signed a formal memorandum of understanding and in 2008, Andriacchi and a team of students from the departments of mechanical engineering and functional restoration began working with BMVSS, to develop an alternative knee joint. The Stanford team recounted, “They came to us and said, ‘We need a really cheap knee joint for our clinic that can be produced locally for roughly \$35 per unit.’...We started thinking...about the technical aspects of the solution. But then we...figured out that they already had a knee joint they were using that cost about \$25. So this wasn't an unmet need.”⁵ Joel Stadler, one of the students on the team, explained, “We had to look at the problem from more than just a mechanical or engineering perspective. There was a psychological issue for these amputees when they walked on these really cheap knee joints. It boiled down to the fact that they were really scary to use. So the problem...started to evolve from simply designing a cheap knee joint to helping the user feel more confident walking.”⁶

The Stanford team faced the challenge of not having direct access to their end user. According to Stadler, “If we had a question about how the joint would make contact with the leg, we could answer that with an amputee in the U.S...But if the question had anything to do with how the product was going to be culturally accepted, then that was much harder to figure out here.”⁷ To overcome this and to appreciate the exact needs of the end users, the Stanford team watched videos of amputees in India and relied on the first-hand knowledge and expertise of the BMVSS team. Eventually, the Stanford team got hold of a specialized device they could strap on to a bent knee to replicate the experience of walking with a prosthesis.⁸

In addition, both the Stanford and BMVSS teams had to bear in mind that the knee they devised should be one that BMVSS could easily fabricate without depending on external manufacturer who could raise prices on a whim or delay shipping. Hence, the knee needed to use locally available materials and a simple manufacturing process. Once the initial design was ready, Neukermans funded 14 visits by the Stanford team to BMVSS’s Jaipur center, so that the two teams could collaboratively work on developing the technology and conduct field trials. The Stanford and BMVSS teams finally developed the Stanford-Jaipur Knee, a device that cost \$20, and consisted of five pieces, made of oil-filled nylon polymer that self-lubricated with use, and four standard fasteners (see **Exhibit 6**). The design allowed for 165-degree range of motion, allowing the person to kneel and squat. A titanium replacement could cost as much as \$10,000.⁹ By early 2013, BMVSS had fitted close to 5,000 Stanford-Jaipur knees based on this prototype.

Other Innovations

Initially, making the stump socket by creating a mold of the patient’s leg required expensive casting material. However, the BMVSS team devised a simple low-cost process to replicate this. They used medical gauze bandages that cost \$0.10 per roll and plaster of Paris to make a mold of the patient’s leg. Technicians filled this mold with plaster of Paris to make an exact replica of the patient’s leg. Molten HDPE sheets were vacuum-fitted over this replica to create the total contact socket for the prosthesis. BMVSS used technology adopted by most international prosthetics manufacturers to ensure the best possible alignment between the socket and the patient’s leg.⁸ The most expensive equipment used in the fabrication and fitment process was the vacuum-creating machine that cost approximately \$4,000.

The BMVSS team constantly sought to improve the Jaipur limbs and was always looking for feedback. Veerendra Raj Mehta, president, BMVSS, Delhi branch, recalled, “There was a problem with the screw connecting the foot to the shank. Fortuitously, a visiting professor noticed this and suggested a change in the screw’s shape. This helped increase the foot’s life by six months.” D.R. Mehta chimed in, “Our research is need-based and involves experts, who are very critical, as well as those with ground level experience. In fact, the process of consultation, irrespective of the person’s expertise, gives us good ideas. Based on the principles of Kaizen, our method has yielded more successful results than several other institutions in the field of prosthetics and orthotics.” Despite these successes, critics often described BMVSS’s innovation model as unsystematic and unstructured.

The Indian Space Research Organization (ISRO) had agreed to share its polyurethane technology (used in insulation and shock absorption) with BMVSS. Use of the polyurethane polymer to fabricate the Jaipur Foot was expected to reduce the cost by 40% and weight by 60%, while also reducing fabricating time. The polyurethane foot was under trial and had demonstrated higher abrasion

⁸The alignment process ensured normal gait and reduced pressure between the patient’s limb and total contact socket.

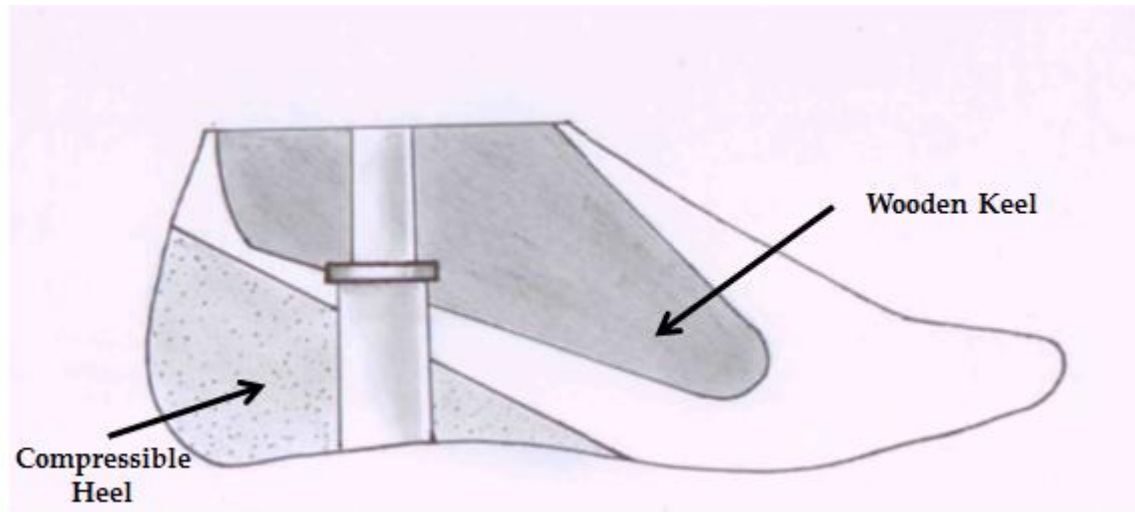
resistance and slip resistance^h compared to the traditional Jaipur Foot. However, the life of the polyurethane foot was one year as compared to the Jaipur Foot's life of approximately three and half years. BMVSS and several organizations including Dow and Massachusetts Institute of Technology (MIT) were involved in research to increase the longevity of the polyurethane foot.

Furthering its collaboration with Stanford University, BMVSS sought to develop artificial hands that would allow patients to write, lift objects, drive and eat with greater ease than the available low-cost models. BMVSS was also collaborating with MIT to develop an all-terrain wheelchair.

In 2005, each Jaipur Foot, including the shank and brace, cost BMVSS approximately \$30; comparable cost for a prosthetic limb in the United States was around \$8,000.¹⁰ By 2012, the cost of the Jaipur foot was approximately \$45.¹¹ This was much lower than the cost of modern prostheses, which could range from \$10,000 to \$100,000, depending on the design and material.¹² Despite its low cost, the Jaipur Foot was functionally capable of providing the same range of movement as some of the more expensive prosthetic limbs. In fact, Sudha Chandra, a dancer and well-known Indian television and film artiste, had used the Jaipur Foot through her 20-year career and given numerous dance performances while wearing the Jaipur limb.

^hAbrasion resistance allowed materials to resist wear, while slip resistance prevented slippage or skidding on smooth surfaces.

Exhibit 1 Diagrammatic Representation of the SACH Foot



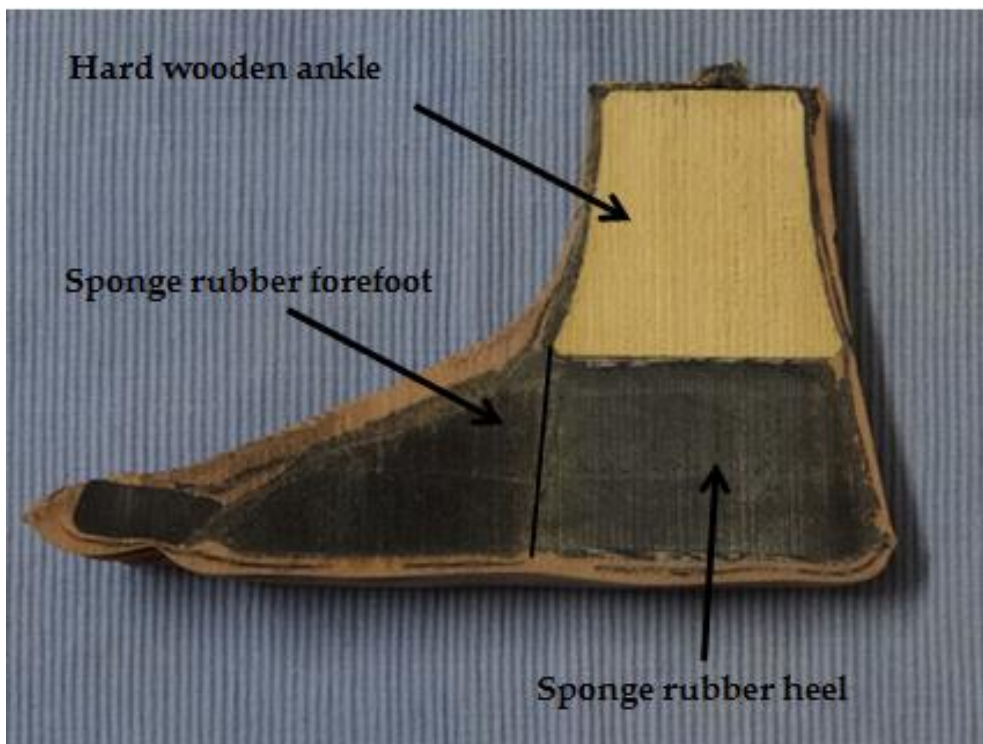
Source: Casewriter.

Exhibit 2 Jaipur Limb for Below-the-knee and Above-the-knee Amputees



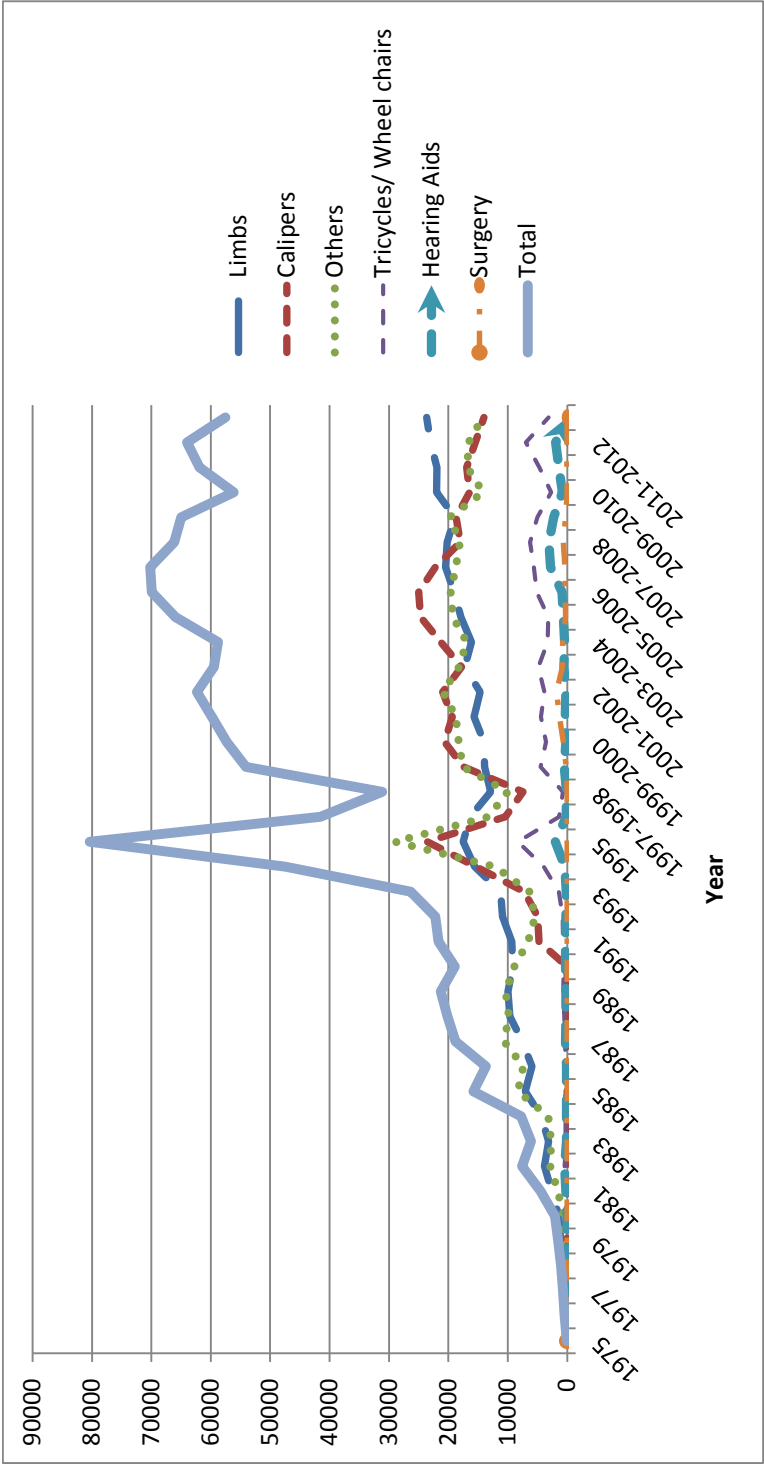
Source: Company documents.

Exhibit 3 Anatomy of the Jaipur Foot



Source: Company documents.

Exhibit 4 Breakdown of Assistance Provided by BMVSS (1975-2013)



Source: Company documents.

Note: From 1975 to 1995, assistance provided was tabulated for each calendar year, after which, starting March 1996, assistance provided was tabulated for each financial year (April 1 to March 31). In this graph, assistance provided during January 1, 1996, to March 31, 1996, has been included in the period 1996-1997, after which assistance provided is tabulated according to each financial year.

Exhibit 5 List of Major Donors (2008-2011)

2009-2009		2009-2010		2010-2011		2011-2012	
Donor	Amount (in \$)	Donor	Amount (in \$)	Donor	Amount (in \$)	Donor	Amount (in \$)
Government of India	875,740	Government of India	400,000	Government of India	863,660	Government of India	764,000
Dow Chemical International	200,000	Dow Chemical International	200,000	Dow Chemical International	200,000	Glen Mark Pharmaceuticals Ltd.	180,000
Paul Hamlyn Foundation	64,000	Paul Hamlyn Foundation	64,000	Paul Hamlyn Foundation	64,000	Indian Railway Catering Tourism Corporation Ltd.	40,617
Lupin Industries	50,000	Union Bank of India	40,000	Andhra Bank	40,000	Rotary Club Peenya Bangalore	13,000
Union Bank of India	20,000	Jiv Daya Foundation	23,608	Glen Mark Pharmaceuticals Ltd.	90,000	Rail Tel Corporation of India	42,238
		S. Foundation	10,000			Ratan Lal Kanwar Lal	10,767
		Ratan Lal Kanwar Lal	17,440			Raghu Singh Mala Mathur Charity Trust	22,000
		Arvind Construction Co.	40,000			Manav Sewa Sanidhi	23,854
		Glen Mark Pharmaceuticals Ltd.	30,000			Hemendra Kothari Foundation	20,000
						State Trading Corporation	90,000
						Vaibhav Medical & Education Foundation	10,000
						Kampani Charitable Trust	10,000
						J.M. Financial Foundation (Mumbai)	10,000
						Clear Trail Technologies	10,000
						Poly Medicure Ltd.	10,000
						Security Printing and Minting Corporation of India Ltd.	20,000
						SMS Samiti	19,509
	1,209,740		825,048		1,257,660		1,295,984

Source: Company documents.

Exhibit 6 Stanford-Jaipur Knee

Source: Company documents.

Endnotes

¹ Tim McGirk, "The \$28 foot," *TIME*, October 1, 1997, <http://www.time.com/time/subscriber/article/0,33009,987107-2,00.html>, accessed July 2013.

² Vulcanization was a process of strengthening rubber by combining it with sulphur and other additives and then applying heat and pressure.

³ Amrit Dhillon, "Steps to freedom," *South China Morning Post*, March 30, 2009, <http://www.scmp.com/article/675127/steps-freedom>, accessed July 2013.

⁴ Amrit Dhillon, "Steps to freedom," *South China Morning Post*, March 30, 2009, <http://www.scmp.com/article/675127/steps-freedom>, accessed July 2013.

⁵ "The Jaipur Knee Project I: Getting the Need Right," August 3, 2012, post on blog "Global Health Innovation," <http://stanfordglobalhealth.com/2012/08/03/the-jaipur-knee-project-i-getting-the-need-right/>, accessed July 2013.

⁶ Ibid.

⁷ Ibid.

⁸ Ibid.

⁹ "The 50 Best Inventions of 2009," *TIME*, November 12, 2009, http://www.time.com/time/specials/packages/article/0,28804,1934027_1934003_1933963,00.html, accessed July 2013.

¹⁰ C.K. Prahlad, *The Fortune at the Bottom of the Pyramid*, (New Delhi: Pearson Education, 2005), p. 244.

¹¹ Mark Magnier, "India group's prosthetic device a boon for thousands of amputees," *Los Angeles Times*, February 8, 2012, <http://articles.latimes.com/2012/feb/08/world/la-fg-india-prosthesis-20120209>, accessed July 2013.

¹² "The Jaipur Knee Project I: Getting the Need Right," August 3, 2012, post on blog "Global Health Innovation," <http://stanfordglobalhealth.com/2012/08/03/the-jaipur-knee-project-i-getting-the-need-right/>, accessed July 2013.